Physics II — Key Definitions & Memory Tricks

Phy 4143 · Lecture 1–3 · Circuit basics, Ohm's Law, KCL/KVL, node/junction/branch, sources, transformations

Basic Quantities

Voltage

Electrical potential difference between two points in a circuit; the force that drives current. Measured in volts (V).

■ ***Memory trick:*** *Think: Voltage = water PRESSURE pushing water (current) through a pipe. No pressure, no push.*

Current

The flow of electric charge (electrons) through a conductor. Measured in amperes (A).

■ ***Memory trick:*** *Current = the actual FLOW of water in the pipe. "I" for "In motion."*

Resistance

How much a material blocks the flow of current. Measured in ohms (Ω).

■ ***Memory trick:*** *Resistance = a NARROW pipe — restricts flow. Thin pipe (high R) means less water gets through.*

Ohm's Law

V = I × R — voltage is directly proportional to current and inversely proportional to resistance.

■ ***Memory trick:*** *Magic Triangle (V on top, I and R below): cover what you want — the other two show the formula.*

Circuit Fundamentals & Laws

Electric Circuit

A closed path for transmitting current through electrical and magnetic fields.

■ ***Memory trick:*** *Like a race track — current only "runs the race" if the track is a closed loop.*

KCL (Kirchhoff's Current Law)

The algebraic sum of currents at any node/junction is zero — current in = current out.

■ ***Memory trick:*** *Roundabout rule: cars entering a roundabout = cars leaving it. Nothing disappears.*

KVL (Kirchhoff's Voltage Law)

The algebraic sum of voltages around any closed loop is zero.

■ ***Memory trick:*** *Hiking loop trail: total elevation UP = total elevation DOWN when you return to the start.*

Node, Junction, Branch, Loop, Mesh

Node

A point where two or more circuit elements are joined.

■ ***Memory trick:*** *N for “Neighbors meet” — any 2+ wires touching.*

Junction

A point where three or more circuit elements meet; current divides here — this is where KCL applies.

■ ***Memory trick:*** *Traffic JUNCTION — roads SPLIT into multiple directions.*

Branch

The part of a network that lies between two junction points.

■ ***Memory trick:*** *Tree BRANCH — grows between two forks (junctions).*

Loop

Any closed path in a network.

■ ***Memory trick:*** *A circular running track — you start and end at the same point.*

Mesh

The smallest loop that cannot be divided further.

■ ***Memory trick:*** *Window MESH screen — the smallest square “hole” in the grid.*

Sources

Source

A device that converts mechanical, chemical or thermal energy into electrical energy; an active network element.

■ ***Memory trick:*** *Source = a PUMP — converts other energy into “pressure” (voltage) to push charge.*

Voltage Source

A two-terminal device whose voltage is constant and independent of the current drawn from it (ideal: zero internal resistance).

■ ***Memory trick:*** *A STUBBORN pump — keeps the SAME pressure no matter how much water you draw.*

Current Source

A two-terminal device that supplies the same current regardless of load voltage (ideal: infinite internal resistance).

■ ***Memory trick:*** *A STUBBORN hose — always sprays the SAME flow, no matter what’s attached.*

Energy-Storing Elements

Inductor

A coil that stores energy in a magnetic field when current flows through it. v = L × (di/dt). Measured in henries (H).

■ ***Memory trick:*** *Inductor = a heavy FLYWHEEL — resists any sudden CHANGE in current, just like a flywheel resists a sudden change in spin speed.*

Capacitor

Two conductive plates that store energy in an electric field. i = C × (dv/dt). Measured in farads (F).

■ ***Memory trick:*** *Capacitor = a water TANK — stores charge like a tank stores water; resists any sudden CHANGE in voltage (water level).*

Resistor Connections & Dividers

Series Connection

Resistors are in series when the same current flows through all of them. Req = R1 + R2 + …

■ ***Memory trick:*** *Single-lane road — every car (electron) passes through at the same rate.*

Parallel Connection

Resistors are in parallel when they share the same two nodes, so the same voltage appears across all of them.

■ ***Memory trick:*** *Multiple lanes, same start and end — same "pressure" (voltage) across every lane.*

Voltage Divider

v_R2 = [R2 / (R1+R2+R3)] × Vs — splits total voltage among series resistors, proportional to resistance.

■ ***Memory trick:*** *The BIGGER resistor grabs the BIGGER voltage share.*

Current Divider

i_R2 = [(1/R2) / (1/R1+1/R2+1/R3)] × Is — splits total current among parallel resistors.

■ ***Memory trick:*** *Current takes the EASY path — the SMALLER resistor gets MORE current.*

Transformations

Source Transformation

A practical voltage source (V, series R) can be replaced by an equivalent current source (I=V/R, R in parallel), and vice versa.

■ ***Memory trick:*** *Swap trick: V with R in SERIES becomes I=V/R with the SAME R in PARALLEL.*

Y-Δ / Δ-Y Transformation

Converts a 3-terminal Y (star) resistor network into an equivalent Δ (triangle) network, or vice versa — used when a circuit isn't plain series/parallel.

■ ***Memory trick:*** *Y looks like a STAR, Δ looks like a TRIANGLE — swap shapes to simplify tricky circuits.*